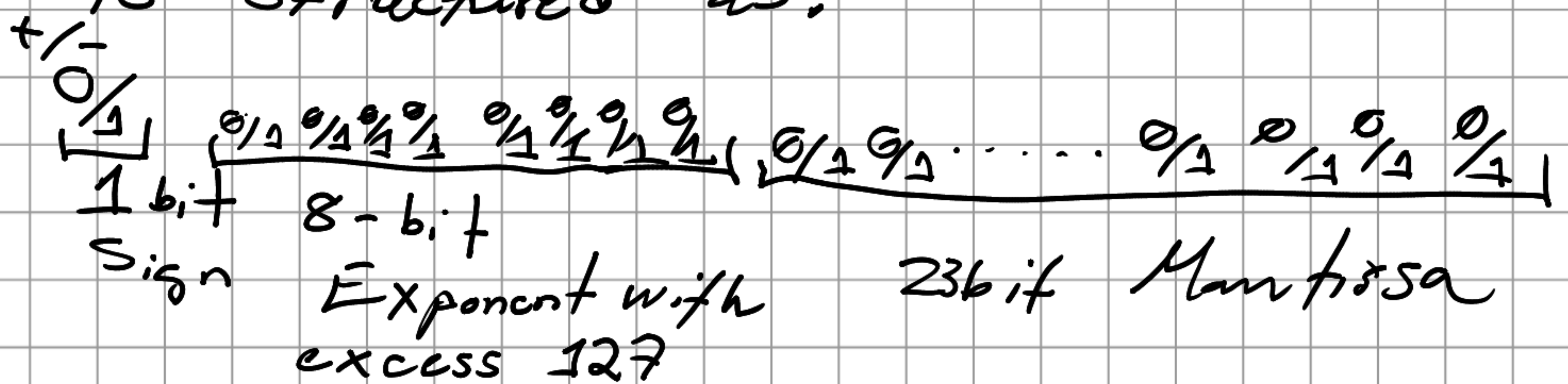


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IEEE 754 to decimal conversion.

The conversion is exactly what has been seen in class on how to analyze and convert binary integer to decimal.

The only key is to know as we saw last week (Feb 27, 29/2024), that IEEE 754 for 32 bit floating point numbers, according to Chapter 5 of the Textbook, Englander, Irv et al, is structured as:



The 23 bit Mantissa is normalized and the structure is $1.0 \frac{0}{2} \frac{0}{2} \dots \frac{0}{2}$

where the integer left of the radix point is never represented.

Thus all Mantissa bits are the 0's/1's (0's or 1's) after the radix (binary) point.

Convert IEEE 754 to Decimal. The number was truncated and not rounded

$\underbrace{0}_{\text{Sign}} \underbrace{10001000}_{\text{Exponent (8 bit)}} \underbrace{0111000011011100101011}_{\text{Mantissa (23 bits)}}$
 $\rightarrow$ + the number is positive

Exponent 8 bits: 10001000

Mantissa: 23 bits: 0111000011011100101011

Convert the exponent.

$$\begin{array}{c}
 \text{7} \quad \text{6} \quad \text{5} \quad \text{4} \quad \text{3} \quad \text{2} \quad \text{1} \quad \text{0} \\
 \hline
 2 \quad (1) \quad 0 \quad 0 \quad 0 \quad (1) \quad 0 \quad 0
 \end{array}
 = 1 \times 2^7 + 1 \times 2^3 = 1 \times 2^1 (2^3)^2 + 1 \times 2^3 =$$

$$= 1 \times 2 \cdot (8) + 1 \times 8$$

This is because:

$$1 \times 2^3 = \frac{2}{1} \times \frac{2}{2} \times \frac{2}{3} = 4 \times 2 = 8$$

$$\text{Now: } 2^7 = 1 \times 2^1 (8)^2 = 1 \times 2 (64) = 128$$

$$\text{So } 10001000 = 1 \times 2^7 + 1 \times 2^3 = 1 \times 2(8^2) + 1 \times 8 = 128 + 8 = 136$$

Now the exponent is in excess 127 notation.

$$\text{So: } \underbrace{10001000}_{\text{Exp.}} = 136 = 127 + X$$

$$X = 136 - 127 = 9$$

Now we begin to patch up our number

$$1 \cdot \underbrace{01110000110111100101011}_{\text{Mantissa}} \times 2^{\uparrow}_{\text{Exponent}}$$

→ This 1 is the implicit integer that is not included in the mantissa in IEEE 754!

So

$$1.0 \times 2^9 + 0.01110000110111100101011 \times 2^9 = (1.0 + 0.01110000110111100101011) \times 2^9$$

n =	0	-1	-2	-3	-4	-5	-6	-7	-8	-9	-10	-11	-12	-13	-14	-15	-16	-17	-18	-19	-20	-21	-22	-23
2 ×	1	0	1	1	1	0	0	0	0	1	1	0	1	1	1	1	0	0	1	0	1	0	1	1
	1	1	1	1	1	0	0	0	0	1	1	0	1	1	1	1	0	0	1	0	1	0	1	1
	1	1	1	1	1	0	0	0	0	1	1	0	1	1	1	1	0	0	1	0	1	0	1	1
	1	1	1	1	1	0	0	0	0	1	1	0	1	1	1	1	0	0	1	0	1	0	1	1
	1	1	1	1	1	0	0	0	0	1	1	0	1	1	1	1	0	0	1	0	1	0	1	1

$$= \frac{0}{2^1} + \frac{1}{2^2} + \frac{1}{2^3} + \frac{1}{2^4} + \frac{0}{2^5} + \frac{0}{2^6} + \frac{0}{2^7} + \frac{0}{2^8} + \frac{1}{2^9} + \frac{1}{2^{10}} + \frac{0}{2^{11}} + \frac{1}{2^{12}} + \frac{1}{2^{13}} + \frac{1}{2^{14}} + \frac{1}{2^{15}} + \frac{0}{2^{16}} + \frac{0}{2^{17}} + \frac{1}{2^{18}} + \frac{0}{2^{19}} + \frac{1}{2^{20}} + \frac{0}{2^{21}} + \frac{1}{2^{22}} + \frac{1}{2^{23}} =$$

$$= \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{0}{32} + \frac{0}{64} + \frac{0}{128} + \frac{0}{256} + \frac{1}{512} + \frac{1}{1024} + \frac{0}{2048} + \frac{1}{4096} + \frac{1}{8192} + \frac{1}{16384} + \frac{1}{32768} + \dots$$

$$\dots + \frac{0}{65,536} + \frac{0}{131,072} + \frac{1}{262,144} + \frac{0}{524,288} + \frac{1}{1,048,576} + \frac{0}{2,097,152} + \frac{1}{4,194,304} + \frac{1}{8,388,608} =$$

$$= \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{512} + \frac{1}{1024} + \frac{1}{4096} + \frac{1}{8192} + \frac{1}{16,384} + \frac{1}{32,768} + \frac{1}{262,144} + \frac{1}{1,048,576} + \frac{1}{4,194,304} + \frac{1}{8,388,608} =$$

Fractional Part of Mantissa.

$$= \frac{0}{2^1} + \frac{1}{2^2} + \frac{1}{2^3} + \frac{1}{2^4} + \frac{0}{2^5} + \frac{0}{2^6} + \frac{0}{2^7} + \frac{0}{2^8} + \frac{1}{2^9} + \frac{1}{2^{10}} + \frac{0}{2^{11}} + \frac{1}{2^{12}} + \frac{1}{2^{13}} + \frac{1}{2^{14}} + \frac{1}{2^{15}} + \frac{0}{2^{16}} + \frac{0}{2^{17}} + \frac{1}{2^{18}} + \frac{0}{2^{19}} + \frac{1}{2^{20}} + \frac{1}{2^{21}} + \frac{1}{2^{22}} + \frac{1}{2^{23}} =$$

$$= \frac{1}{2^2} + \frac{1}{2^3} + \frac{1}{2^4} + \frac{1}{2^9} + \frac{1}{2^{10}} + \frac{1}{2^{12}} + \frac{1}{2^{13}} + \frac{1}{2^{14}} + \frac{1}{2^{15}} + \frac{1}{2^{18}} + \frac{1}{2^{20}} + \frac{1}{2^{22}} + \frac{1}{2^{23}}$$

$$= \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{512} + \frac{1}{1024} + \frac{1}{4096} + \frac{1}{8192} + \frac{1}{16384} + \frac{1}{32768} + \frac{1}{262144} + \frac{1}{1048576}$$

$$+ \frac{1}{4194304} + \frac{1}{8388608} =$$

14 DECIMAL digits of precision

BINARY

0.25 00000000000000	0.010000000000000000000000000000
0.125 00000000000000	0.001000000000000000000000000000
0.0625 00000000000000	0.000100000000000000000000000000
0.001953125 00000000	0.000000000100000000000000000000
0.0009765625 00000000	0.000000000010000000000000000000
0.000244140625 00000000	0.000000000000100000000000000000
0.0001220703125 00000000	0.000000000000010000000000000000
0.00006103515625 00000000	0.000000000000001000000000000000
0.000030517578125 00000000	0.000000000000000100000000000000
0.000003814697265625	0.000000000000000010000000000000
0.00000095367431640625	0.000000000000000001000000000000
0.000000238418579101562	0.000000000000000000100000000000
0.000000011920928755078	0.000000000000000000010000000000

$$= 0.440892577171326 000000$$

$$+ 1.0000000000000000000000$$

$$= 1.440892577171326 000000 \rightarrow \text{Mantissa}$$

$$\times 2^9$$

But if exponent is a power of 2...
it's easier just to apply bit shifts.

BINARY

0.01000000000000000000000000000000	$= 2^{-2}$
0.00100000000000000000000000000000	$= 2^{-3}$
0.00010000000000000000000000000000	$= 2^{-4}$
0.00000000010000000000000000000000	$= 2^{-9}$
0.00000000001000000000000000000000	$= 2^{-10}$
0.00000000000010000000000000000000	$= 2^{-12}$
0.00000000000001000000000000000000	$= 2^{-13}$
0.00000000000000100000000000000000	$= 2^{-14}$
0.00000000000000001000000000000000	$= 2^{-15}$
0.00000000000000000010000000000000	$= 2^{-16}$
0.00000000000000000000000000000000	$= 2^{-20}$
0.00000000000000000000000000000000	$= 2^{-22}$
0.00000000000000000000000000000000	$= 2^{-23}$

$\times 2^9$

Left shift 9 bits

We push our bits string
9 bits to the left

Mantissa

0.01000000000000000000000000000000
0.00100000000000000000000000000000
0.00010000000000000000000000000000
0.00000000010000000000000000000000
0.00000000001000000000000000000000
0.00000000000010000000000000000000
0.00000000000001000000000000000000
0.00000000000000100000000000000000
0.00000000000000001000000000000000
0.00000000000000000010000000000000
0.00000000000000000000000000000000
0.00000000000000000000000000000000
0.00000000000000000000000000000000
0.00000000000000000000000000000000
0.00000000000000000000000000000000

$= 2^{-2} \times 2^9 = 2^7 = 128$
$= 2^{-3} \times 2^9 = 2^6 = 64$
$= 2^{-4} \times 2^9 = 2^5 = 32$
$= 2^{-9} \times 2^9 = 2^0 = 1$
$= 2^{-10} \times 2^9 = 2^{-1} = \frac{1}{2}$
$= 2^{-12} \times 2^9 = 2^{-3} = \frac{1}{2^3} = \frac{1}{8}$
$= 2^{-13} \times 2^9 = 2^{-4} = \frac{1}{2^4} = \frac{1}{16}$
$= 2^{-14} \times 2^9 = 2^{-5} = \frac{1}{2^5} = \frac{1}{32}$
$= 2^{-15} \times 2^9 = 2^{-6} = \frac{1}{2^6} = \frac{1}{64}$
$= 2^{-16} \times 2^9 = 2^{-7} = \frac{1}{2^7} = \frac{1}{128}$
$= 2^{-20} \times 2^9 = 2^{-11} = \frac{1}{2^{11}} = \frac{1}{2048}$
$= 2^{-22} \times 2^9 = 2^{-13} = \frac{1}{2^{13}} = \frac{1}{8192}$
$= 2^{-23} \times 2^9 = 2^{-14} = \frac{1}{2^{14}} = \frac{1}{16384}$

Please refer to the spreadsheet
for the final result.